

PSP RDS 19c upgrade plan

SYS

	Milestone	ETA (mm/dd/yyyy)	Status	Notes
1	Goldengate 19c upgrade	06/19/2020		
2	Upgrade GoldenGate from 12c 19c			
3	Test replication			
4	Parallel database setup for upgrade - Monolith	07/03/2020		
5	create a parallel 12c database by cloning current database			1 new RDS
6	Setup and configure GG replication 12c 12c			1 new EC2. Used single EC2 to configure goldengate replication for monolith and ibob databases.
7	Setup and configure Veridata			
8	data validation via Veridata compare			
9	upgrade target created rds and configure replication 12c 19c			
10	data validation via Veridata compare			
11	setup monitoring for GG replication			
12	Parallel database setup for upgrade - IBOB	07/03/2020		
13	create a parallel 12c database by cloning current database			1 new RDS. This also holds veridata repository which has comparison data.
14	Setup and configure GG replication 12c 12c			1 new EC2.Used single EC2 to configure goldengate replication for monolith and ibob databases.
15	data validation via Veridata compare			
16	upgrade target created rds and configure replication 12c 19c			
17	data validation via Veridata compare			
18	setup monitoring for GG replication			
19	Capture 12c performance baseline metrics	07/17/2020		Dependency on application/RTB team
20	Application regression and sanity with Oracle 12c			RTB team to run testing
21	capture application and db performance metrics in 12c			RTB team to capture application performance metrics
22	data validation via Veridata compare			
23	Capture 19c performance metrics and verify 19c 12c replication	07/31/2020		Dependency on application/RTB team
24	Application regression and sanity with Oracle 19c			RTB team to switch to oracle 19c and run regression testing
25	capture application and db performance metrics in 19c and review			RTB team to capture application performance metrics
26	data validation via Veridata			

Production

	Milestone	ETA (mm/dd/yyyy)	Status	Notes
1	Goldengate 19c upgrade	07/24/2020		
2	Upgrade GoldenGate from 12c 19c			
3	Test replication			

4	Parallel database setup for upgrade - Monolith	09/04/2020		
5	setup new 12c database by cloning current prod database			1 new RDS. We would need to maintain this till cutover.
6	Setup and configure GG replication 12c 12c (new RDS)			1 new EC2. We would need to maintain this till cutover.
7	data validation via Veridata compare			
8	upgrade target created rds and configure replication 12c 19c			
9	data validation via Veridata compare			
10	create 19c read replica			1 new RDS. We can build this 1 week before cutover.
11	set up monitoring for database and GG replication			
12	Parallel database setup for upgrade - IB OB	09/04/2020		
13	setup new 12c database by cloning current prod database			1 new RDS. We would need to maintain this till cutover.
14	Setup and configure GG replication 12c 12c (new RDS)			1 new EC2. We would need to maintain this till cutover.
15	data validation via Veridata compare			
16	upgrade target created rds and configure replication 12c 19c			
17	data validation via Veridata compare			
18	set up monitoring for database and GG replication			
19	Switchover to 19c	10/01/2020		Dependency on application team. We are planning to switch application to 19c on 09/11/2020. Cooling period would be 2 weeks.
20	Switchover to 19c & wait for 1 week			
21	data validation at regular intervals via Veridata			
22	Upgrade DR database - Monolith	10/23/2020		
23	upgrade DR database for monolith to 19c			
24	configure replication			
25	data validation via Veridata compare			
26	set up monitoring for database and GG replication			
27	Upgrade DR database - IBOB	10/23/2020		
28	upgrade DR database for ibob to 19c			
29	configure replication			
30	data validation via Veridata compare			
31	set up monitoring for database and GG replication			
32	Decommission 12c infra including RDS and EC2 instances	11/06/2020		

Note: We have added buffer time and all dates calculated based on positive scenarios. We also have dependency on application team which we have highlighted and hence dates might get move accordingly. If there are any issues related to upgrade, GoldenGate, or Veridata, we will try to fix and highlight immediately in case if it's taking long time to fix.